C1.3 Photosynthesis

NAME	DATE	TOTAL
		_____ / 42

About 45 minutes, or one section at a time. A calculator is allowed. The number of marks shows how many separate points to make.

Section A: Recall

8 MARKS

1 WRITE the word equation for photosynthesis. [C1.3.2] [2]

2 STATE the source of the oxygen released in photosynthesis. [C1.3.3] [1]

3 STATE the two colours of light that chlorophyll absorbs best. [C1.3.5] [2]

4 LIST three factors that can limit the rate of photosynthesis. [C1.3.7] [3]

Section B: Understanding

14 MARKS

5 EXPLAIN the role of hydrogen from water in photosynthesis. [C1.3.2] [2]

6 EXPLAIN how light energy is transformed when a pigment absorbs it. [C1.3.5] [3]

7 DISTINGUISH between an absorption spectrum and an action spectrum. [C1.3.6] [3]

8 EXPLAIN what is meant by a limiting factor, using light intensity as an example. [C1.3.7] [3]

9 **COMPARE** greenhouse and FACE experiments as ways of predicting future photosynthesis. [C1.3.8] [3]

Section C: Working with data

14 MARKS

All data in this section are invented for practice. The patterns are realistic.

10 [C1.3.4]

A student ran a leaf extract on a thin-layer strip. The solvent moved 75 mm.

SPOT	COLOUR	DISTANCE MOVED / MM
1	Orange-yellow	72
2	Yellow	48
3	Blue-green	30
4	Yellow-green	18

a **CALCULATE** Rf for spot 3. [2]

b **IDENTIFY** the pigment in spot 3, by its colour. [1]

c **SUGGEST** why the spots separate. [1]

11 [C1.3.6, C1.3.7]

Students counted bubbles from pondweed in 5 minutes, with a lamp at different distances.

DISTANCE OF LAMP / CM	10	20	30	40	50
BUBBLES IN 5 MIN	60	60	40	22	14

a CALCULATE the rate at 30 cm, in bubbles per minute. [1]

b DESCRIBE the relationship. [2]

c EXPLAIN why the rate does not rise between 20 cm and 10 cm. [2]

d SUGGEST two improvements to this method. [2]

e DESCRIBE how you could vary the concentration of CO_2 in this experiment. [1]

f The experiment was repeated with coloured filters. PREDICT the colour that would give the lowest rate, giving a reason. [2]

Section D: Longer answer

6 MARKS

12 EXPLAIN how the energy in sunlight comes to be stored in glucose. [C1.3.1–5] [6]